

- New RAS (Reliability, Availability and Serviceability) features
- Enhanced security features including something called Privileged Access Never state, that prevents unauthorised access to user data

The 32-bit ARM architecture is supported by a number of operating systems including all major mobile platforms like Android, iOS, Windows 10 mobile, Chrome OS, and Firefox OS.

Many desktop Linux distributions including Gentoo, Ubuntu, BSD and OpenSolaris are also supported.

Most of the current generation SoCs used in mobiles are ARMv8 based including the Apple A-x series, Samsung's Exynos series and Qualcomm's Snapdragon series.

Intel x86

Intel, founded by the famous Gordon Moore and Robert Noyce, is the world's largest chip manufacturer. They're also the inventor and proprietor of one of the most common and ubiquitous architecture, x86 and microprocessors based on it.

The x86 architecture encompasses the whole family of CPUs by Intel. Getting its name from Intel's breakthrough 8086 CPU, it was initially developed for embedded systems and small single-user personal computers. More than 40 years later, the x86 architecture is everywhere, powering laptops, netbooks, personal computer, servers, midrange workstations, portable devices and even supercomputers.

The main difference between the x86 architecture and the rest on the list is that x86 is CISC based compared to RISC architecture on which the rest are based. Generally, CISC based processors are more powerful but not efficient (check the CISC vs RISC box for more information).

x86 based Intel SoCs for mobile devices started coming out in 2012. Intel announced its partnership with Google to extend support for its Android OS on its x86 platform, in the same year.

Intel tried to compete with ARM architecture based SoCs with its Atom series. The chip gained initial attraction but after a couple of phones from ASUS that definitely had the potential, enjoyed mild sales. Intel realised that it couldn't compete with ARM even if it enjoyed the status of being one of biggest technological company, at least not in the mobile market segment.

So, in 2016 Intel decided to exit the smartphone market and cancelled its next generation of x86 based SoCs codenamed the SoFIA and Broxton.